whoisit Instagram Relationship Checker

Full Audit, Remediation, and Open-Source Benchmark

Audited version: original 1.0 archive supplied by the user **Corrected version:** 2.2.0 **Audit date:** 30 September 2026 **Scope:** Chrome/Edge extension, standalone dashboard, relationship comparison, JSON import, privacy, resilience, accessibility, performance, and packaging

1. Executive result

The original archive was visually promising but its primary connection path was not deployable as written: a normal website cannot call a browser extension with an empty extension ID, and the extension did not declare an external connection mechanism or website bridge. Stop requests did not stop Instagram fetching, progress updates were discarded, completion timers leaked, and live requests lacked timeout, retry, duplicate-cursor, and useful rate-limit handling.

Versions 2.0.0 through 2.2.0 replace that fragile flow with:

- a bundled dashboard that opens directly from the extension and needs no extension ID;
- a restricted content-script bridge for `cybroatrix.com`, `www.cybroatrix.com`, localhost, and `127.0.0.1`;
- job-based start, progress, polling, completion, error, and cancellation messages;
- conservative request pacing, timeouts, limited retry for transient server errors, explicit 429 handling, and cursor-loop protection;
- adaptive relationship-list passes that stop on a reported-total match or one no-growth retry, with clearly marked partial results when Instagram withholds accounts;
- a pre-analysis follow gate for `@cybroatrix`, enforced in the dashboard and again in the Instagram page worker;
- an official Instagram JSON-export workflow that works entirely offline;
- three exact result categories: does not follow you back, you do not follow back, and mutuals;
- responsive light/dark UI, persistent theme selection, whoisit logo, community links, keyboard-friendly controls, visible progress, search, sorting, result pagination, profile links, CSV export, and JSON export;
- a local development server whose documented `/website/` route is covered by a smoke and regression check;
- dependency-free runtime, deterministic tests, static checks, and a repeatable benchmark.

2. Original defects and fixes

Severity	Original defect	Impact	Version 2.2 remediation
Critical	Website called `chrome.runtime.sendMessage()` with an empty extension ID	The website could not reliably connect to an unpacked or published extension	Bundled extension dashboard plus allowlisted website bridge
Critical	Live scan trusted one apparently complete cursor walk	Instagram page reordering silently omitted 72 followers and 21 following in observed validation, creating false positives	Union adaptive repeated passes; mark incomplete comparisons as partial; keep their exports disabled

Severity	Original defect	Impact	Version 2.2 remediation
High	No membership gate existed before live analysis	Live scans could begin without verifying the product's required community follow	Exact target lookup plus `/friendships/show/<id>/`; dashboard remains locked and worker independently refuses analysis until verified
High	Missing Instagram profile totals were coerced to zero	A later live test displayed 450 of 0 followers and 463 of 0 following, making a usable capture appear invalid	Support `user` and `data.user`, recover profile-page counts, preserve unknown totals as null, and use no-growth convergence
High	Incomplete scans repeated identical passes and hid all output	A 79/78 followers and 15/27 following scan finished in 1:02 but showed no relationships; longer retries could not recover Instagram-withheld accounts	Stop after one no-growth retry; cap passes at three/two; show partial categories with an explicit accuracy warning
High	Stop only rejected the background wait	Instagram requests continued after the user pressed Stop	Per-job `AbortController`; stop reaches the Instagram content worker
High	Progress messages were received and discarded	Dashboard appeared frozen during long scans	Stored job progress with dashboard polling and processed counts
High	No request timeout or 429-specific behavior	Scans could hang or repeatedly stress a rate-limited account	25-second request timeout, clear 429 stop message, limited 5xx/network retry
High	Pagination cursor was not checked for repetition	A bad Instagram response could create a long request loop	Duplicate-cursor set and 5,000-page hard guard
Medium	Analysis timer interval was never cleared on success/error	Background work leaked after each scan	Job lifecycle has terminal complete/error/stopped states
Medium	Global stop flag allowed concurrent scans to interfere	One scan could stop or corrupt another	Unique job IDs and one active Instagram-tab job at a time
Medium	Connection fallback treated any page cookie as login	False connection status and unnecessary script permission	Session endpoint check; removed `scripting` permission and injection fallback
Medium	Only fragile private endpoints were supported	Product stopped working whenever live endpoints changed	Official export JSON mode is a reliable local fallback
Medium	Only CSV and two categories were available	Limited analysis and result portability	CSV + JSON; non-followers, fans, and mutuals
Low	Large result sets rendered in one pass	Slow DOM rendering for high-count accounts	200-row incremental rendering with Load more
Low	Inline single-file dashboard was hard to test and maintain	UI and logic regressions were easy to introduce	Separated HTML/CSS/core/app and synchronized extension build

3. Correctness definition

For normalized, unique username sets:

- **Does not follow you back** = Following minus Followers
- **You do not follow back** = Followers minus Following
- **Mutuals** = Following intersect Followers

The implementation uses `Set` membership, giving expected time complexity $O(n + m)$ and expected memory complexity $O(n + m)$. Usernames are normalized to lowercase and duplicate export entries are removed before comparison.

4. Test and security results

Check	Result
Core/contract/simulated-worker automated tests	21 passed, 0 failed

Check	Result
100k followers + 100k following exactness test	Passed; 50k non-followers, 50k fans, 50k mutuals
JavaScript syntax checks	12 files passed
Manifest validation	MV3, `tabs` + `storage` only, Instagram-only host access
Cookie API check	No `chrome.cookies` permission or usage
Dynamic code check	No `eval`
Unsafe HTML insertion check	No `innerHTML` assignment in application scripts
Dashboard source/build parity	Passed for JS, CSS, configuration, core logic, and logo asset
Dependency audit	0 known npm vulnerabilities; no runtime dependencies
Original ZIP integrity	Passed decompression test
Local HTTP smoke test	`/website/`, logo, and application script returned successfully

Test boundary

The comparison engine, JSON parser, exports, build synchronization, message contracts, manifest permissions, simulated reordered pagination, unknown-total handling, partial-result delivery, follow-gate wiring, theme controls, branding, extension icons, and synthetic performance were tested locally. User-provided validation showed the 2.0.1 scanner captured 381 of 453 followers and 444 of 465 following; screenshots confirmed at least one false positive. Version 2.0.2 recovered 450 followers and 463 following but exposed unavailable totals as zero. Version 2.0.3 fixed the count response, then a small-account scan showed Instagram returning only 15 of 27 following accounts and the UI hiding all output after 1:02. Version 2.0.4 stops after one no-growth retry and presents that output as partial. A fresh authenticated live scan of 2.2.0, including the follow-status endpoint, must still be run by the user because the audit environment does not use or retain account cookies. Official export mode remains the exact fallback.

5. Synthetic performance benchmark

Runtime: Node.js v24.19.0, Linux x64. Values are medians from warm repeated runs in this audit environment and will vary by device.

Followers	Following	Median comparison	Expected result
1,000	1,000	0.68 ms	500 / 500 / 500
10,000	10,000	7.21 ms	5,000 / 5,000 / 5,000
100,000	100,000	85.04 ms	50,000 / 50,000 / 50,000

The three result numbers are, in order: does not follow you back / you do not follow back / mutuals. Rendering is separately limited to 200 rows per batch so a correct fast comparison does not produce a slow browser page.

6. Open-source comparison

This is a capability and result-delivery benchmark based on each project's public documentation as checked on 28 September 2026. It is not a live account speed race: the tools use different inputs (private web endpoints, official exports, or Instaloader), and no shared authenticated account was supplied.

Project	Input method	How it gives results	Strongest capability	Main trade-off
whoisit 2.2.0	Live browser session after follow verification **and** official JSON export	Verified results when complete; fast, visibly marked partial results when Instagram withholds accounts	Strong balance of usable live output, transparent confidence, dark/light UI, and exact offline fallback	Partial live categories can contain false positives and cannot be exported; live mode requires following the configured account
UNKchr/ig-analyzer	Tampermonkey + Instagram internal REST with GraphQL fallback	Instagram overlay with many tabs, history, target audit, and contextual CSV	Broadest live-analysis and historical feature set	More complex; live scraping has rate/action-block risk; custom non-commercial license
Instagram Data Visualizer	Official JSON/HTML export	Cross-platform Electron dashboard with charts and broad account-data exploration	Richest official-export analysis and privacy posture	No instant live scan; larger desktop installation
SilentSnitch	Official follower/following JSON export	Flutter app list of unfollowers with profile links	Simple offline, no-login workflow	Manual exports; narrower relationship/report feature set
instagram-monitor	Instaloader login, Docker, SQLite	Rotating logs, persistent change history, and optional Telegram alerts	Best scheduled monitoring/notification workflow	Requires credentials in environment configuration and technical setup; no polished GUI

Documented evidence

- UNKchr/ig-analyzer documents internal Friendships REST use, legacy GraphQL fallback, history, non-followers, target audits, contextual CSV, and deliberate pacing because of rate-limit/action-block risk: https://github.com/UNKchr/ig-analyzer
- Instagram Data Visualizer documents local processing, zero network requests, JSON and HTML export support, multi-folder input, connections analysis, and a cross-platform Electron dashboard: https://github.com/journalismrelay/instagram-data-visualizer
- SilentSnitch documents local official-export comparison without login, a non-follower list, and manual refresh limitations: https://github.com/dyingpotato890/SilentSnitch/releases
- instagram-monitor documents Instaloader, credential configuration, SQLite persistence, rotating logs, Docker, and Telegram notifications: https://github.com/JustShinobi/instagram-monitor

No competitor source code was copied into whoisit. The benchmark influenced product decisions only: safer pacing, explicit limitations, local export support, progressive results, and clearer privacy communication.

7. Weighted product scorecard

Scoring: 0 = absent, 3 = present with limitations, 5 = strong. Weights prioritize privacy and dependable results over feature count. Scores are documentation-based and should be rechecked as projects change.

Criterion	Weight	whoisit 2.2.0	ig-analyzer	Data Visualizer	SilentSnitch	instagram-monitor
Privacy / credential safety	20	19	16	20	20	9
Result correctness + clarity	20	19	18	18	14	16
Resilience to Instagram changes	15	13	10	14	12	8
Setup / usability	15	13	10	10	12	5

Criterion	Weight	whoisit 2.2.0	ig-analyzer	Data Visualizer	SilentSnitch	instagram-monitor
Analysis breadth	10	7	10	10	5	8
Export / portability	10	10	8	7	3	6
History / monitoring	10	1	10	4	4	10
Total / 100	**100**	**82**	**82**	**83**	**70**	**62**

Interpretation: whoisit now competes strongly on privacy, low setup friction, clear relationship results, and dual live/offline input. The clearest remaining gap is opt-in historical comparison. Instagram Data Visualizer leads for broad export exploration; ig-analyzer leads for live feature depth; instagram-monitor leads for unattended history/notifications.

8. Recommended next release

1. Add opt-in encrypted local snapshots so users can see new followers, lost followers, and username changes without a server.
2. Add a small fixture-driven browser test suite (Playwright) in CI for dashboard tab switching, JSON upload, search, sorting, pagination, and export downloads.
3. Publish the extension so the production origin and release ID can be tested as a stable pair.
4. Test live mode with a dedicated low-risk Instagram account before public release, beginning with small lists and respecting back-off messages.
5. Add a visible scan-frequency reminder for accounts with thousands of connections.

9. Final assessment

The corrected build is suitable for local testing and controlled beta use. Official export mode is the dependable production path. Live mode now verifies the configured follow requirement before analysis, distinguishes unknown totals from zero, ends stalled scans quickly, and labels incomplete categories as partial rather than hiding all output. Partial live results can contain false positives, so exports stay disabled until verification succeeds. Because live mode and follow verification rely on undocumented endpoints, 2.2.0 still requires a fresh user-side validation.